

Constructions

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Propositional coherence

At all times we maintain a list of prime formulas that have appeared up until now. And at each time, the set of worlds is all possible boolean valuations of these formulas. Before each round of trading, we check which new prime formulas traders will bet on, or have appeared in D_n , and add them to the list, further splitting worlds.

We only have prices and pay-offs for worlds. Traders can keep thinking in terms of arbitrary formulas, but now buying a share of φ will actually mean buying a share of all worlds in which it's satisfied, at the corresponding price:

$$\mathbb{P}_n(\varphi) := \sum_{\mathbb{W} \models \varphi} \mathbb{P}_n(\mathbb{W})$$

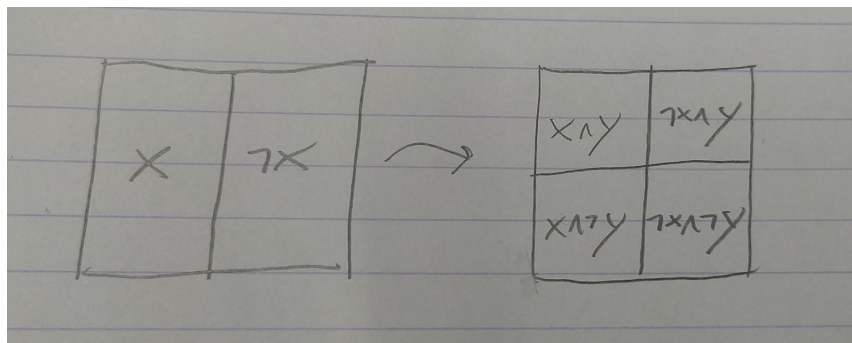
Consider we add to our list the new prime formula φ_p , so that any of our worlds $\mathbb{W}$ gets further split into $\mathbb{W} + \varphi_p$ and $\mathbb{W} + \neg\varphi_p$. If at least one of these worlds is traded on by one of our traders, **MarketMaker** will set its price (and, if only one was traded on, we can deduce the price of the other by subtracting from $\mathbb{P}_n(\mathbb{W})$). Otherwise (if φ_p has just appeared in D_n but not been traded on), we can just set $\mathbb{P}_{n+1}(\mathbb{W} + \varphi_p) = \mathbb{P}_{n+1}(\mathbb{W} + \neg\varphi_p) = 0.5 \cdot \mathbb{P}_n(\mathbb{W})$.

Also, if any trader had one share of $\mathbb{W}$, they automatically get replaced by one share of $\mathbb{W} + \varphi_p$ and one of $\mathbb{W} + \neg\varphi_p$.

Now we'll define a way for worlds to pay off such that formulas also pay off correctly. What we want, intuitively, is the expected payoff from φ to equal the probability of φ , so that prices truly represent probabilities. For this, and since worlds get increasingly more split, we define intermediate payoffs that end up adding up as necessary.

Say $\mathbb{W}$ values n prime formulas (as do all other worlds our inductor is currently tracking). Say also that k of its assignments have been confirmed correct, while $n - k$ are still to be confirmed or refuted. When a trader first comes into possession of one of its shares, it receives as immediate payoff $1/2^{n-k}$. If, on the contrary, one of its assignments has already been refuted (so that we know $\mathbb{W}$ is false), it receives 0. Of course, if the trader sells a share, it immediately loses these same amounts. Now, suppose a trader is in possession of one share, and one further assignment from $\mathbb{W}$ is confirmed true. Then the trader immediately receives $1/2^{n-k-1} - 1/2^{n-k}$ (so that the total payoff it has obtained now equals $1/2^{n-k-1}$). On the contrary, if one assignment is refuted, it immediately loses $1/2^{n-k}$ (so that the total payoff it has obtained now equals 0).

When all prime formulas are eventually decided by the deductive process, it's easy to see the expected reward works out. Consider for example buying a share of x by buying a share of $x \wedge y$ and one of $x \wedge \neg y$. Each of those immediately give $1/4$, so in total $1/2$. If x is refuted, that is lost. If x is confirmed (before y is confirmed or refuted), then each of them returns $1/2 - 1/4$, for a total of 1 . And from now on the total will always be 1 , since even if y is confirmed or refuted, one of the shares will make the trader lose $1/2$, and the other provide $1/2$.

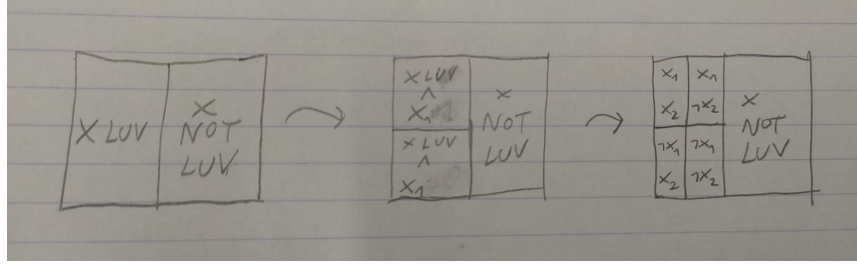


But we need all prime formulas to be eventually decided for this idea to work. Otherwise, consider buying a share of $x \wedge y$. If that is confirmed true, the reward will be 1 . But if, for example, x is true and y undecidable, it will provide a reward of $1/2$, messing up our expected reward, so that we have to take into account the chances of undecidability.

LUV coherence

We introduce a new syntactic feature to the above idea, but it's pretty straight-forward. When a world assigns true to a prime formula of the form $\varphi_x := \exists!x(X(x)) \wedge \exists x(X(x) \wedge 0 \leq x \leq 1)$ (that is, that world believes x to be a $[0, 1]$ -LUV), we split this world with new boolean variables $x_1, \dots, x_n$, where n is the number of the current trading round. x_i represents "the i th binary digit of x is 1". We then treat these variables as the truth values of all other prime formulas.

Notice that now some worlds will be more "fine-grained" (have more variables) than others, but the calculations still work out (provided, again, that all prime formulas are eventually decided). Consider for example buying " x is a $[0, 1]$ -LUV, and furthermore x_1 is true". But we are in round 2, so we buy one share where x_2 is true, and one where it's false. Upon buying those, we immediately obtain $1/8$ from each, that is, $1/4$ in total. Then it is revealed that, if x is indeed a $[0, 1]$ -LUV, then its first digit must be a 1. Thus each one of them provides $1/4 - 1/8$, and so in total my payoff is $1/2$. Finally, when x is confirmed a $[0, 1]$ -LUV, I get the additional $1/2$, and when x_2 is decided, my payoff is held constant. Combinations of the kind "either x is not a $[0, 1]$ -LUV, or it is and x_1 " also work out (it's just then the intermediate rewards might not be powers of $1/2$).



Such a distribution over LUV values gives us some of the nice expectation properties we wanted, within $1/2^n$ precision. Mainly, since

$$\mathbb{P}_n(x > p) = \sum_{\mathbb{W} \models x > p} \mathbb{P}_n(\mathbb{W})$$

we get

$$\mathbb{E}_n(x) = \sum_{\mathbb{W}} \mathbb{W}_n(x) \cdot \mathbb{P}_n(\mathbb{W})$$

where $\mathbb{W}_n(x)$ is calculated in the obvious way, and the n represents its granularity¹.

We can also syntactically enforce more constraints to get "more coherent" distributions, in the sense of taking advantage of the information about LUVs already present. For example, say a world satisfies not only φ_x and φ_y , but also $\exists x, y(X(x) \wedge Y(y) \wedge x = y + r)$, for some $r \in [0, 1] \cap \mathbb{Q}$. Then, we can force this world to reflect this on its estimates of x and y (up to this round's granularity). That is, we just eliminate from our list of worlds those that satisfy φ_x, φ_y , this new sentence, and don't satisfy the constraint.

We can proceed similarly for $x = y \cdot r$ and other nested combinations with more LUVs. In fact, we can simply demand it for any combination using LUVs, rational numbers, sums, products and max that the model believes, similarly to expressible features. The worry here is that this enforces too much structure, but after all these checks will only be carried out for consistency statements within the support of the inductor (even if not yet decided by D_n), so a growing finite amount.

Another worry is that we're only fixing this syntactically, so there will be "clearly equivalent" or "not-clearly, but equivalent" constraints to these ones, that nonetheless don't trigger our checks. This will eventually be arbitrated by traders, so those worlds will go down to low probabilities.

This does mean that we won't get, at every (or any) finite round, something like "all information believed in this world is used in the LUV assessments". But of course, that was impossible in principle: deciding in every finite round whether an existential sentence is equivalent to one of our syntactic constraints amounts already to logical omniscience, since we can deduce truth-values of arbitrary formulas in that way.

What we do get is a way to read off coherent assignments given some "official" channels of information (our syntactic constraints), and the LI will ever-better inform these by its accumulated logical

¹In fact, this leads to a (purely aesthetic) mismatch, because our LUVs' precision doubles each round, while $\mathbb{E}_n$'s precision grows more slowly. Maybe we want to change the definition of $\mathbb{E}_n$ for this purpose.

knowledge.

There are also some ideas to expand worlds more drastically, by having them be models (in the Model Theory sense) in which the LUVs are points (belonging to certain classes of a certain granularity). But there seem to be some problems with how much we compute from these models. For example, all of these models would be finite, but the real models of a theory might be infinite.

Bayesian logical induction

As a side note before moving forward, since $\mathbb{Q}$ will be defined in terms of worlds, it will have finitely many prime formulas in its support, but we'll be able to calculate the probabilities of arbitrary compositions of those formulas, just by calculating in which worlds they are satisfied². So it makes sense also for $\mathbb{P}$ to work like that.

But actually, it's not clear which sentences $\mathbb{P}_{n-1}$ should have an opinion on. Certainly on boolean combinations of prime formulas in $\mathbb{Q}_{n-1}$ and all the prime formulas of the form $\mathbb{Q}_n = \mathcal{Q}$. But it seems like this would imply some knowledge about many other formulas: since we don't know which prime formulas will be added to $\mathbb{Q}_n$'s support, any prime formula can appear in them (with different probabilities). So this already implies some expected information about these formulas that $\mathbb{P}_{n-1}$, as a bayesian reasoner, allegedly should implement. We could hope to have something like "these probabilities for any additional formula marginalize to 0.5", which is the "impartial guess" on these sentences $\mathbb{P}_{n-1}$ could have. Unfortunately, the LI framework treats formulas outside the support as having probability 0, not 0.5. And I could try and change that, but actually I feel like that's there for a good reason: the alternative 0.5 guess seems bad, and will break coherence properties mentioned above unless we just use logical omniscience to check all infinitely many prime formulas for the correct values.

What I'm trying to say more generally is that $\mathbb{P}_{n-1}$ thinking about all possible $\mathbb{Q}_n$ forces it to have opinions about all formulas, and I don't know how to deal with that. This makes me consider maybe $\mathbb{P}_{n-1}$ shouldn't consider all possible $\mathbb{Q}_n$. But it's not enough to make $\mathcal{Q}$ "more coarse-grained": this will only be solved if the number of prime formulas possibly appearing in $\mathbb{Q}_n$ is finite. Can we get away with having $\mathbb{P}_{n-1}$ only talk about prime formulas up to length $f(n)$, or something like that? What additional structure does this give me I can use though? Well, only having to correct finitely many coherence constraints, so that I can do it in finite time...

In fact, the situation is even worse (that is, the definition we're looking for has even stronger consequences). Having a sensible definition for $\mathbb{P}_{n-1}(\mathbb{Q}_n = \mathcal{Q})$ gives us almost everything we want, since then we could go

$$\mathbb{P}_{n-1}(\mathbb{Q}_{n+1} = \mathcal{Q}') := \sum_{\mathcal{Q}} \mathbb{P}_n^{\mathcal{Q}}(\mathbb{Q}_{n+1} = \mathcal{Q}') \cdot \mathbb{P}_n(\mathbb{Q}_n = \mathcal{Q})$$

²I worried that this might somehow give traders in $\mathbb{Q}$ (or the $\mathbb{P}$ process) arbitrary access to the outputs of finite computations, since we can reduce any computational problem to a SAT problem, and then just query whether that formula is satisfiable, by checking whether it has positive probability. The way to check that would be to plug in prime formulas (with non-0, non-1 probabilities) in our support as the independent boolean variables of the SAT problem, and so if the probability of the overall formula is non-0, some assignment must satisfy it. In any event, if this is not worrying it's because the longer the computation we want to decide, the more independent boolean variables we need on the SAT problem. So this is capped by the amount of prime formulas in the support of $\mathbb{Q}_n$. I do think this does mean that, late enough, $\mathbb{Q}_n$ will have decided all computations shorter than a certain $f(n)$. But this already happened by the traders getting more computation.

and so on recursively, where $\mathbb{P}_n^Q$ is calculated as if $\mathbb{Q}_n$ were Q . Actually, it might be the case this is not possible, because our construction doesn't just take $\mathbb{Q}_{<n}$ and $\mathbb{Q}_n$ as separate inputs, but I don't see right now how that could look like. Most notably, since the bayesian 100% trusts Q , probably $\mathbb{P}_n$ only depends on $\mathbb{Q}_n$, and not previous states.

But maybe that's exactly what we're getting wrong: maybe we need our observer to extrapolate from past states, instead of blindly adopting the current state (that is, not having 100% belief on the current state). In fact, my previous idea of "keeping track of the previous convergence of prices" sounds like a particular case of this.

Actually, here's an intuition pump for why we'd want that: consider the sentence

$$\varphi := "Q_n(\varphi) = 0.5"$$

Assume, for example, $\mathbb{Q}_n(\varphi) = 0.7$. If blindly adopting it, our observer would say $\mathbb{P}_n("Q_n(\varphi) = 0.5") = 0.7$. But of course, if it carried out the slightest bit of analysis on that sentence's structure, it would notice that it has just observed $\mathbb{Q}_n(\varphi) = 0.7$, and so the sentence is false, and assign $\mathbb{P}_n(\varphi) = 0$. That is, instead of blindly adopting what it's seen, it also updates on the fact that it's seen it, jumping up a meta-level.³

Again, we can only define this syntactically, so what about equivalents to the prime formulas " $\mathbb{Q}_n(\varphi) = p$ "? Well, we could try to propagate these meta-updates through entailments we already have guesses for. For example, suppose $\neg\varphi \rightarrow \psi$ is also in $\mathbb{Q}_n$'s support. Then, since we know $\neg\varphi$ true, we'd want

$$\mathbb{P}_n(\psi) = \mathbb{P}_n(\neg\varphi \rightarrow \psi)$$

We could start with $\mathbb{P}_n(\neg\varphi \rightarrow \psi)$ being just $\mathbb{Q}_n(\neg\varphi \rightarrow \psi)$, and then iteratively apply all of these corrections in some privileged order. We'll find a fix-point in the finite support.

A general statement of this procedure could be: "given $\mathbb{P}_n$ already knows a formula true / false, check that all formulas that are equivalent because of this have the same probabilities".

Probably the neat way of doing it is considering worlds. But do I just share the resulting probability equally between the other worlds? Sounds good.

³Now, if $\mathbb{P}$ also updated on observing itself, we might run into self-reference problems. Consider the sentence

$$\varphi := "P_n(\varphi) = 0"$$

and suppose it's in the support of $\mathbb{Q}_n$. Then it would seem like we won't find a fix-point for $\mathbb{P}_n$. But this is not necessary: it can just update on observing Q and not itself.